第5章 热量传递基础

「热传导的基本定律」

一维稳态热传导过程：$q=-\lambda\frac{dt}{dx}$ ⇒ 热通量矢量 = −比例系数 × 温度梯度

导热系数：$\lambda=\lambda_0(1+kt)$

「平壁内的热传导」 $Q=\frac{\lambda A\Delta t}{b}$ 稳态平板法公式

对于热传导面积 A 的平壁，热量传递速率为

$$Q=\frac{\Delta t}{\frac{b}{\lambda A}}=\frac{\Delta t}{R}$$，其中热阻 $\boxed{R=\frac{b}{\lambda A}}$ ⇒ n层平壁，$Q=\frac{t_1-t_{n+1}}{\sum R_i}$

☆

「圆筒壁内的热传导」

$Q=\frac{t_1-t_2}{R}$，其中 $R=\frac{b}{\lambda A_m}$，$b=r_2-r_1$，$A_m=2\pi L\frac{r_2-r_1}{\ln\frac{r_2}{r_1}}$

⇒ n层圆筒壁：$Q=\frac{t_1-t_{n+1}}{\sum_{i=1}^{n}\frac{b_i}{\lambda_i A_{mi}}}$

「球壳壁内的热传导」

$$Q=\frac{t_1-t_2}{\frac{1}{4\pi\lambda}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)}$$

(※)沸腾传热的三个区域：

- 自然对流区
- 核状沸腾区 ⇒ 工程主用
- 膜状沸腾区

「牛顿冷却定律」

当主体温度为 t_f 的流体被温度为 t_w 的壁面加热或冷却时，

传热面上的热通量为 $q=\alpha(t_w-t_f)$ or $\alpha(t_f-t_w)$

$$\begin{cases} Nu=\frac{\alpha L}{\lambda} \\ Re=\frac{du\rho}{\mu} \\ Pr=\frac{C_p\mu}{\lambda} \end{cases}$$

$Gr=\frac{g\beta\Delta t L^3\rho^2}{\mu^2}$

「对流传热系数」

非物性，为传递性质

$\alpha=f(u,L,\mu,\lambda,\rho,c_p,g\beta\Delta t)$

「流体在圆形直管内作湍流时的对流传热系数」

迪图斯－贝尔特公式：$\begin{cases} Nu=0.023Re^{0.8}Pr^n \\ \text{or}\ \alpha=0.023\frac{\lambda}{d}\left(\frac{du\rho}{\mu}\right)^{0.8}\left(\frac{C_p\mu}{\lambda}\right)^n \end{cases}$

☆

当流体被加热，$n=0.4$；流体被冷却，$n=0.3$

流体定性温度 $t_m=\frac{t_进+t_出}{2}$

「流体在圆形直管内作层流时的对流传热系数」

当 $Gr<25000$（$Re<2300$）时，忽略自然对流影响

当 $Gr>25000$ 时，$\begin{cases} Nu=1.86Re^{\frac{1}{3}}Pr^{\frac{1}{3}}\left(\frac{d}{L}\right)^{\frac{1}{3}}\left(\frac{\mu}{\mu_w}\right)^{0.14} \\ \alpha'=0.8(1+0.015Gr^{\frac{1}{3}})\alpha \end{cases}$（不要求）

「流体在圆形弯管内的流动」

$\alpha'=(1+1.77\frac{d}{R})\alpha$

R：曲率半径

「集总参数法响应公式」

$$\frac{t-t_\infty}{t_0-t_\infty}=\exp\left(-\frac{\alpha A}{\rho c V}\tau\right)$$

适用条件：$Bi=\frac{\alpha L_c}{\lambda}<0.1$

「有相变传热」：冷凝传热

「冷凝传热」

- 膜状冷凝
- 滴状冷凝

$\alpha_膜<\alpha_滴$

其中液膜是主要热阻

膜状冷凝传热过程强化：

- 正确的蒸汽流动方向
- 过热蒸汽
- 及时排放不凝性气体

在整个管束上的平均对流传热系数计算：

$$\alpha = \frac{\sum_{i=1}^{n} \alpha_i A_i}{\sum_{i=1}^{n} A_i}$$

[流体在管壳间的对流传热]

$$Nu = \frac{\alpha d_e}{\lambda} = 0.36 Re^{0.55} Pr^{\frac{1}{3}} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$ X

管子按正方形排列 $d_e = \frac{4(t^2 - \frac{\pi}{4}d_o^2)}{\pi d_o}$

正三角形排列 $d_e = \frac{4(\frac{\sqrt{3}}{2}t^2 - \frac{\pi}{4}d_o^2)}{\pi d_o}$ X

[冷凝传热]

纯蒸气在竖壁上进行层流膜状冷凝时的对流传热系数

$$\alpha = 1.13 \left[\frac{r g \rho^2 \lambda^3}{\mu L (t_s - t_w)}\right]^{\frac{1}{4}}$$

湍流 $\longrightarrow$ $\alpha^* = \alpha\left(\frac{\mu^2}{\lambda^3 \rho^2 g}\right)^{\frac{1}{3}} = 0.0077 Re_L^{0.4}$

水平单管和管束外的冷凝传热：

$$\alpha = 0.725 \left[\frac{r g \rho^2 \lambda^3}{\mu d (t_s - t_w)}\right]^{\frac{1}{4}}$$

[沸腾传热]

[自然对流传热]

$$Nu = C(Gr Pr)^n$$

or $\alpha = C\frac{\lambda}{L}\left(\frac{\rho^2 g \beta \Delta t L^3}{\mu^2} \frac{C_p \mu}{\lambda}\right)^n$

[斯蒂芬－玻尔兹曼定律]

$$E_b = C_0\left(\frac{T}{100}\right)^4$$

黑体辐射系数 $C_0 = 5.67\ W/(m^2 \cdot K^4)$

灰体辐射系数 C

物体的黑度 or 发射率 $\varepsilon = \frac{E}{E_b} = \frac{C}{C_0}$

$\Rightarrow$ 灰体的辐射能为 $E = \varepsilon C_0\left(\frac{T}{100}\right)^4$

[克希霍夫定律]

$$\alpha = \varepsilon = \frac{E}{E_b}$$

在热平衡时，灰体的吸收率在数值上等于同温度下该物体的黑度

化原Ⅱ 第5章习题

[5-1]

管道外半径 $r_1=\frac{d_1}{2}=25\,mm=0.025\,m$

矿渣棉 $\delta_1=40\,mm$, $\lambda_1=0.11\,W/(m\cdot°C)$

煤灰泡沫砖 $\delta_2=45\,mm$, $\lambda_2=0.12\,W/(m\cdot°C)$

管道外表面 $t_1=400°C$，泡沫砖外表面 $t_3=50°C$

$$\begin{cases} r_1=0.025\,m \\ r_2=0.065\,m \\ r_3=0.11\,m \end{cases}$$

$$R_i=\frac{b_i}{\lambda A_m}=\frac{(r_{i+1}-r_i)\ln\frac{r_{i+1}}{r_i}}{\lambda\times 2\pi \cancel{L}(r_{i+1}-r_i)}=\frac{\ln\frac{r_{i+1}}{r_i}}{2\pi\lambda_i}$$

则 $\begin{cases} R_1=1.3825\,(m\cdot°C)/W \\ R_2=0.6978\,(m\cdot°C)/W \end{cases}$

$\Sigma R=R_1+R_2=2.0803\,(m\cdot°C)/W$

则 $\frac{Q}{L}=\frac{\Delta t}{\Sigma R}=168.25\,W/m$

则 $t_2=t_1-\frac{Q}{L}\cdot R_1=167.4°C$

[变式5-1]

$$\begin{cases} r_1=\cancel{0.0015\,m}\ 0.03\,m \\ r_2=0.08\,m \\ r_3=0.12\,m \end{cases}$$

$t_1=250°C$, $t_3=30°C$

(1) $R_i=\frac{\cancel{r_{i+1}-r_i}\ln\frac{r_{i+1}}{r_i}}{\lambda\times 2\pi}$

则 $\begin{cases} R_1=1.9513\,(m\cdot°C)/W \\ R_2=0.4964\,(m\cdot°C)/W \end{cases}$

$\Sigma R=R_1+R_2=2.4477\,(m\cdot°C)/W$

则 $\frac{Q}{L}=\frac{t_1-t_3}{\Sigma R}=89.88\,W/m$

(2)

$t_2=t_1-\frac{Q}{L}R_1=74.62°C$

[5-2]

$r_1=0.0015\,m$

$R=2.22\times10^{-3}\,\Omega/m$

$\cancel{r_3}\ r_2=0.0025\,m$

$R_1=\frac{\ln\frac{r_2}{r_1}}{2\pi\lambda}=0.5420\,(m\cdot°C)/W$

则 $\frac{Q}{L}=\frac{t_2-t_1}{R_1}=119.93\,W/m$

令 $\frac{Q}{L}=I^2R \Rightarrow I_m=232.4\,A$

[变式5-2]

$r_1=0.001\,m$

$r_2=0.0018\,m$, $\lambda=0.20\,W/(m\cdot°C)$

$\frac{R}{L}=5.0\times10^{-3}\,\Omega/m$

$t_1=70°C$, $t_2=25°C$

$R_1=\frac{\ln r_2-\ln r_1}{2\pi\lambda}=0.4677\,(m\cdot°C)/W$

$\frac{Q}{L}=\frac{t_1-t_2}{R_1}=96.22\,W/m=\frac{RI^2}{L}$

$\Rightarrow I=138.7\,A$

[5-3] $A=0.02\,m^2$, $b=0.01\,m$

$\lambda=\frac{Q\cdot b}{A\Delta t}=\frac{IVb}{A\Delta t}$

$\Rightarrow \begin{cases} \lambda_1=\cancel{98}\ 0.98\,W/(m\cdot°C) \\ \lambda_2=0.8664\,W/(m\cdot°C) \end{cases}$

$\lambda_{avg}=\frac{\lambda_1+\lambda_2}{2}=0.9232\,W/(m\cdot°C)$

而 $\begin{cases} \lambda_1=\lambda_0(1+k\cdot t_{avg,1}) \\ \lambda_2=\lambda_0(1+k\cdot t_{avg,2}) \end{cases}$

$\Rightarrow \lambda=0.6771\times(1+0.002237t)\,W/(m\cdot°C)$

[变式5-3]

$A=0.015\,m^2$, $b=0.02\,m$

$\lambda=\frac{Q\cdot b}{A\Delta t} \Rightarrow \begin{cases} \lambda_1=3.73\,W/(m\cdot°C) \\ \lambda_2=2.6\,W/(m\cdot°C) \end{cases}$

$\cancel{\lambda_{avg}=}$

$\Rightarrow \begin{cases} \lambda_0=0.3400 \\ k=0.02659 \end{cases}$

[5-4]

(1) $r_1 = 0.027m$, $r_2 = 0.03m$, $r_3 = 0.06m$, $r_4 = 0.09m$

$\lambda_{钢} = 16\ W/(m\cdot°C)$ $\lambda_{石} = 0.16\ W/(m\cdot°C)$, $\lambda_{软} = 0.04\ W/(m\cdot°C)$

$$\begin{cases} R_{钢} = \dfrac{\ln\frac{r_2}{r_1}}{2\pi\lambda_{钢}} = 1.048\times10^{-3}\ (m\cdot°C)/W \\ R_{石} = \dfrac{\ln\frac{r_3}{r_2}}{2\pi\lambda_{石}} = 0.6895 \\ R_{软} = \dfrac{\ln\frac{r_4}{r_3}}{2\pi\lambda_{软}} = 1.6133 \end{cases}$$

$$\frac{Q}{L} = \frac{\Delta t}{\Sigma R} = \frac{-120}{2.304} = -52.08\ W/m$$

(2) $\dfrac{Q'}{L} = -37.95\ W/m$

(3) 自然对流散热量与导热热流量相等

$$\left|\frac{Q}{L}\right| = \alpha\pi d_4 (t_{空} - t_4) \Rightarrow \alpha = 9.21\ W/(m^2\cdot°C)$$

互换后，t_4^* 未知 $\Rightarrow t_4^* = 12.56°C$

$$\frac{Q^*}{L} = 38.76\ W/m$$

(4) 将导热系数大的材料置于外层、导热系数小的材料置于内层，其冷量损失较小.

[变式5-4]

$r_1 = 21mm = 0.021m$, $r_2 = 0.024m$, $r_3 = 0.049m$, $r_4 = 0.074m$

管道管壁热阻可忽略

A: $R_{玻} = \dfrac{\ln\frac{r_3}{r_2}}{2\pi\lambda_{玻}} = 2.524\ (m\cdot°C)/W$ $\dfrac{Q}{L} = \dfrac{\Delta t}{\Sigma R} = -12.74\ W/m$

$R_{聚} = \dfrac{\ln\frac{r_4}{r_3}}{2\pi\lambda_{聚}} = 2.187\ (m\cdot°C)/W$

B: $R'_{玻} = \dfrac{\ln\frac{r_4}{r_3}}{2\pi\lambda_{玻}} = 1.458\ (m\cdot°C)/W$ $\dfrac{Q'}{L} = \dfrac{\Delta t}{\Sigma R'} = -11.44\ W/m$

$R'_{聚} = \dfrac{\ln\frac{r_3}{r_2}}{2\pi\lambda_{聚}} = 3.787\ (m\cdot°C)/W$

内层为导热系数偏小的，冷量损失更少

[5-5]

集总参数法响应公式： $\dfrac{t-t_\infty}{t_0-t_\infty} = e^{-\frac{\alpha A}{\rho c V}\tau}$

适用条件： $Bi = \dfrac{\alpha L_c}{\lambda} \le 0.1$，其中特征长度 $L_c = \dfrac{V}{A}$

$L_c = \dfrac{V}{A} = \dfrac{r}{3} = 3.33\times10^{-5}\ m$

$Bi = 5.83\times10^{-4} < 0.1$ ， $\dfrac{\alpha A}{\rho c V} = 3.088\ s^{-1}$， $\tau_c = \dfrac{1}{3.088} = 0.324s$

则 $\dfrac{t-t_\infty}{t_0-t_\infty} = \exp(-3.088\tau) \Rightarrow t = 200 - 175\times\exp(-3.088\tau)$

[变式5-5]

$$\frac{t-t_\infty}{t_0-t_\infty} = e^{-\frac{\alpha A}{\rho c V}\tau}$$

~~其~~，$Bi = \dfrac{\alpha L_c}{\lambda} = \dfrac{\alpha r}{3\lambda} = 0.03125 < 0.$

则 $t = 60 + 790\times e^{-0.0534\tau}$

$t_{10s} = 523.14\ °C$

$t_{30s} = 219.18\ °C$， $t_{60s} = 92.07°C$

【5-6】

(1) $Bi = \frac{a L_c}{\lambda} = 6.94 \times 10^{-3} < 0.1$，可以

(2) $$t = 50 + 200 e^{-\frac{aA}{\rho c V}\tau}$$

$$= 50 + 200 e^{-0.00266\tau}$$

$$\begin{cases} t_{300s} = 140.05°C \\ t_{600s} = 90.54°C \\ t_{1800s} = 51.67°C \end{cases}$$

【5-7】

半无限大体非稳态导热问题，用误差函数解

$$\frac{T(x,\tau) - T_s}{T_0 - T_s} = erf\left(\frac{x}{2\sqrt{a\tau}}\right)$$，深度 x

热扩散率 $a = \frac{\lambda}{\rho c} = 1.378 \times 10^{-7} m^2/s$

$$\frac{T - T_s}{T_0 - T_s} = \frac{15}{35} = 0.4286 = erf\left(\frac{x}{2\sqrt{a\tau}}\right) \Rightarrow \frac{x}{2\sqrt{a\tau}} = 0.40$$

$$\Rightarrow x = 0.676 m$$

【5-8】 $Nu = f(Gr, Pr)$

【变式 5-6】 $Bi = \frac{a L_c}{\lambda} = 4.22 \times 10^{-4} < 0.1$

$$t = 20 + 280 e^{-\frac{aA}{\rho c V}\tau}$$

$$= 20 + 280 e^{-\frac{1}{270}\tau}$$

令 $t = 100°C$，$\Rightarrow \tau = 338.25s$

【变式 5-7】

$$\frac{T - T_s}{T_0 - T_s} = erf\left(\frac{x}{2\sqrt{a\tau}}\right)$$

$$a = \frac{\lambda}{\rho c} = 6.629 \times 10^{-7} m^2/s$$

$$\Rightarrow T = 3.83°C$$

第6章 传热过程计算与换热器

「热量衡算方程」

$$Q = m_h C_{ph}(t_{h1} - t_{h2}) = m_c C_{pc}(t_{c2} - t_{c1})$$

「传热速率方程」

$$Q = K \Delta t_m A$$

「总传热系数」 $\frac{1}{K} = \frac{1}{\alpha_i} + \frac{b}{\lambda} + \frac{1}{\alpha_o}$

「平均传热温差」 $\Delta t_m = \dfrac{\Delta t_1 - \Delta t_2}{\ln \frac{\Delta t_1}{\Delta t_2}}$

「传热效率」

$$\varepsilon = \frac{Q}{Q_{max}} = \frac{\text{实际传热速率}}{\text{理论最大传热速率}}$$ ☆

其中 $Q_{max} = (mC_p)_{min}(t_{h1} - t_{c1})$

则有 $Q = \varepsilon Q_{max} = \varepsilon (mC_p)_{min}(t_{h1} - t_{c1})$

「冷热流体的传热单元数」

$$\begin{cases} NTU_c = \int_{t_{c1}}^{t_{c2}} \frac{dt_c}{\Delta t} = \frac{t_{c2} - t_{c1}}{\Delta t_m} = \frac{KA}{m_c C_{pc}} \\ NTU_h = \frac{KA}{m_h C_{ph}} \end{cases}$$

而在换热器中 $NTU = \dfrac{KA}{(mC_p)_{min}}$

「传热效率和传热单元数的关系」

$$\varepsilon = \frac{1 - \exp((1-C_R)\cdot NTU)}{C_R - \exp((1-C_R)\cdot NTU)}$$ ，其中 $C_R = \dfrac{(mC_p)_{min}}{(mC_p)_{max}}$ ← 热容量比

「总传热系数与热阻串联模型」

$$\frac{1}{K_o} = \frac{1}{\alpha_o} + R_{fo} + \frac{\delta}{\lambda}\cdot\frac{A_o}{A_m} + R_{fi}\frac{A_o}{A_i} + \frac{1}{\alpha_i}\frac{A_o}{A_i}$$

or $$\frac{1}{KA} = \frac{1}{\alpha_1 A_1} + \frac{R_{s1}}{A_1} + \frac{b}{\lambda A_m} + \frac{R_{s2}}{A_2} + \frac{1}{\alpha_2 A_2}$$

「Dittus - Boelter 关联式」

管内湍流时，$\alpha_i \propto u^{0.8}$

「温差修正系数 F_T」 ×

1-2型：

$$F_T = \frac{\sqrt{R^2+1}\ \ln\frac{1-PR}{1-P}}{(R-1)\ln\dfrac{2-P(1+R-\sqrt{R^2+1})}{2-P(1+R+\sqrt{R^2+1})}}$$

, where $R = \dfrac{T_{h1} - T_{h2}}{T_{c2} - T_{c1}}$, $P = \dfrac{T_{c2} - T_{c1}}{T_{h1} - T_{c1}}$

$$NTU = \int_{T_2}^{T_1} \frac{dT}{T-t}$$

$$dQ = -m_h C_{ph}\, dT = -m_c C_{pc}\, dt = K_x\, dA\ (T-t)$$

对于并流换热器，有

☆ $$\varepsilon = \frac{1 - \exp[-(1+C_R)\cdot NTU]}{1 + C_R}$$

(*) 为强化传热，翅片管换热器的翅片应安装在"对流传热系数小的一侧"

(*) 并联管程数 $\frac{N}{n_p}$

(*) 列管式换热器的分类 $\begin{cases}\text{固定管板式} \\ \text{浮头式} \\ \text{U形管式}\end{cases}$

化原Ⅱ 第6章习题1

[6-1]

解 $\begin{cases} d_o = 25\,mm = 0.025\,m \\ d_i = 0.020\,m \end{cases}$

$d_m = \dfrac{d_o - d_i}{\ln \frac{d_o}{d_i}} = 0.02241\,m$

面积比 $\dfrac{A_o}{A_m} = 1.1156$，$\dfrac{A_o}{A_i} = 1.2500$

则 $\Sigma R = \dfrac{1}{10000} + 0 + \dfrac{0.0025}{45} \times 1.1156 + 1.5\times10^{-3}\times 1.2500 + \dfrac{1}{1000} \times 1.2500$

$= 3.287\times10^{-3}\,m^2\cdot{}^\circ C/W$

则 $K_o = \dfrac{1}{\Sigma R} = 304.2\,W/(m^2\cdot{}^\circ C)$

[变式6-1]

$\begin{cases} d_o = 0.032\,m \\ d_i = 0.026\,m \end{cases} \Rightarrow d_m = \dfrac{d_o - d_i}{\ln \frac{d_o}{d_i}} = 0.029\,m$

则 $\dfrac{1}{K_o} = \dfrac{1}{5000} + 0 + \dfrac{0.003}{45} \times \cancel{1.108}\ 1.104 + 1.0\times10^{-3} \times \cancel{1.515}\ 1.231 + \dfrac{1}{3000} \times \cancel{1.515}\ 1.231$

外表面 $\Rightarrow K_o = \cancel{431.56}\ 522.2\,W/(m^2\cdot{}^\circ C)$

内表面 $\Sigma R_i = \Sigma R_o \times \dfrac{A_i}{A_o} \Rightarrow K_i = 646\,W/(m^2\cdot{}^\circ C)$

[6-2]

$d_o = 0.028\,m$，$d_i = 0.023\,m$，$\lambda = 45\,W/(m\cdot{}^\circ C)$，$d_m = \dfrac{d_o - d_i}{\ln \frac{d_o}{d_i}} = 0.0254\,m$

(1) $\dfrac{a_{i1}}{a_{i2}} = \left(\dfrac{1}{2}\right)^{0.8} = 0.57435$，则 $a_{i1} = 0.57435\,a_{i2}$

$$\begin{cases} \dfrac{1}{K_1} = \dfrac{1}{a_o} + \dfrac{0.0025}{45} \times \dfrac{d_o}{d_m} + \dfrac{1}{0.57435\,a_{i2}} \times \dfrac{d_o}{d_i} \\ \dfrac{1}{K_2} = \dfrac{1}{a_o} + \dfrac{0.0025}{45} \times \dfrac{d_o}{d_m} + \dfrac{1}{a_{i2}} \dfrac{d_o}{d_i} \end{cases} \Rightarrow \begin{cases} a_{i1} = 1752.4\,W/(m^2\cdot{}^\circ C) \\ a_{i2} = 3051.1\,W/(m^2\cdot{}^\circ C) \\ a_o = 12921.4\,W/(m^2\cdot{}^\circ C) \end{cases}$$

(2) 存在管内、管外污垢热阻

[变式6-2]

$d_o = 0.025\,m$，$d_i = 0.020\,m$，$\lambda = 45\,W/(m\cdot{}^\circ C)$

$d_m = \dfrac{d_o - d_i}{\ln \frac{d_o}{d_i}} = 0.0224\,m$，$\dfrac{a_{i2}}{a_{i1}} = \left(\dfrac{1.5}{1}\right)^{0.8} = 1.383$

则 $\begin{cases} \dfrac{1}{K_1} = \dfrac{1}{a_o} + \dfrac{0.005}{45} \times \dfrac{d_o}{d_m} + \dfrac{1}{a_i} \dfrac{d_o}{d_i} \\ \dfrac{1}{K_2} = \dfrac{1}{a_o} + \dfrac{0.005}{45} \times \dfrac{d_o}{d_m} + \dfrac{1}{1.383\,a_i} \dfrac{d_o}{d_i} \end{cases} \Rightarrow \begin{cases} a_o = 2675.9 \\ a_i = 1661.6 \end{cases}$

[6-3]

$A = 16.5\,m^2$，$m_c = 2.5\,kg/s$，$T_{cond} = 80^\circ C$，$T_{c1} = 20^\circ C$，$C_{p,w} = 4180\,J/(kg\cdot{}^\circ C)$

$Q = C_{pw} m_c (T_{c2} - T_{c1}) = 104500\,W$

$\Delta t_1 = 80 - 20 = 60^\circ C$，$\Delta t_2 = 80 - 30 = 50^\circ C$

$\Delta t_m = \dfrac{\Delta t_1 - \Delta t_2}{\ln \frac{\Delta t_1}{\Delta t_2}} = 54.85^\circ C$

$K = \dfrac{Q}{A\Delta t_m} = 115.5\,W/(m^2\cdot{}^\circ C) \Rightarrow K' = 66.73\,W/(m^2\cdot{}^\circ C)$

$R_f = \dfrac{1}{K_{used}} - \dfrac{1}{K_{new}} = 6.33\times10^{-3}\,m^2\cdot{}^\circ C/W$

[变式6-3]

$K = \dfrac{Q}{A\Delta t_m} = \dfrac{m_c C_{p,w} (T_{c2} - T_{c1})}{A\Delta t_m}$

$\begin{cases} \Delta t_{m1} = \dfrac{20}{\ln \frac{85}{65}} = 74.55^\circ C \\ \Delta t_{m2} = \dfrac{13}{\ln \frac{85}{72}} = 78.32^\circ C \end{cases}$

$R_f = \dfrac{1}{K_2} - \dfrac{1}{K_1}$

$= \left(\dfrac{m_c C_{p,w}}{A}\right)^{-1} \times \left(\dfrac{1}{\frac{T_{c2}-T_{c1}}{\Delta t_{m2}}} - \dfrac{1}{\frac{T_{c2}-T_{c1}}{\Delta t_{m1}}} \right)$

$= 3.664\times10^{-3}\,(m^2\cdot{}^\circ C)/W$

[6-4]

柴油：$243 \rightarrow 155$

原油：$162 \leftarrow 128$

UA？

U总传热系数

则 $\Delta T_1 = 81^\circ C$，$\Delta T_2 = 27^\circ C$

$\Delta T_{m,逆} = 49.15^\circ C$

并流时，$\dfrac{C_h}{C_c} = \dfrac{T_{c2} - T_{c1}}{T_{h1} - T_{h2}} = 0.3864$

而 $C_h (243 - T_{h2}') = C_c (T_{c2}' - 128) = UA\,\Delta T_{m,并流}$

而 $\Delta T_1 = 115^\circ C$，$\Delta T_2 = T_{h2}' - T_{c2}'$

$UA = \dfrac{Q}{\Delta T_{m,逆}} \Rightarrow T_{h2}' = 167^\circ C$，$T_{c2}' = 157^\circ C$

$\Rightarrow \Delta T_{m,并} = 42.99^\circ C$

[变式6-4]

$300^\circ C \rightarrow 200^\circ C$

$\cancel{\#}\ 240^\circ C \leftarrow 150^\circ C$

$\Delta T_{1,逆} = \dfrac{60 - 50}{\ln \frac{60}{50}} = \cancel{44.81^\circ C}\ 54.85^\circ C$

$\Delta T_{1,并} = 47.6^\circ C$

[6-5]

热支衡算求 T_{h2}

$m_h c_p (T_{h1} - T_{h2}) = m_c c_p (T_{c2} - T_{c1})$

$\Rightarrow T_{h2} = 50°C$

逆流 $80 \to 50$

$30 \leftarrow 10$

$\Delta T_1 = 50°C$, $\Delta T_2 = 40°C$, $\Delta T_m = 44.81°C$

并流 $80 \to 50$

$10 \to 30$

$\Delta T_1 = 70°C$, $\Delta T_2 = 20°C$, $\Delta T_m = 39.91°C$

$$\begin{cases} P = \dfrac{T_{c2} - T_{c1}}{T_{h1} - T_{c1}} = 0.2857 \\ R = \dfrac{T_{h1} - T_{h2}}{T_{c2} - T_{c1}} = 1.5 \end{cases} \Rightarrow |F_T| = \left| \frac{\sqrt{R^2+1} \ln \frac{1-PR}{1-P}}{(R-1) \ln \frac{2-P(1+R-\sqrt{R^2+1})}{2-P(1+R+\sqrt{R^2+1})}} \right| = 0.948$$

$\Delta T_{m,1-2} = F_T \times \Delta T_{m1} = 42.5°C$

[6-6]

$m_w = 300 kg/h = 0.08333 kg/s$, $c_{p,w} = 4180 J/(kg \cdot °C)$

$m_o = 360 kg/h = 0.1000 kg/s$, $c_{p,o} = 2100 J/(kg \cdot °C)$

$T_{h1} = 175°C$, $T_{c1} = 25°C$, $T_{c2} = 90°C$

$Q = m_w c_{p,w} (T_{c2} - T_{c1}) = 22650 W = m_o c_{o,w} (T_{h1} - T_{h2})$

$\Rightarrow T_{h1} = 67.14°C$ —— $175°C \to 67.14°C$

逆流 LMTD: $\begin{cases} \Delta T_1 = 85°C \\ \Delta T_2 = 42.14°C \end{cases}$ $90°C \leftarrow 25°C$

$$\Rightarrow \Delta T_m = \frac{85 - 42.14}{\ln \frac{85}{42.14}} = 61.11°C$$

换热器 1: 单壳程双管程, $K_1 = 625 W/(m^2 \cdot °C)$

$P = 0.4333$, $R = 1.659$

$$P_{max} = \frac{2}{1 + R + \sqrt{R^2+1}} = 0.4351$$

$$F_T = \frac{\sqrt{R^2+1} \ln \frac{1-PR}{1-P}}{(R-1) \ln \frac{2-P(1+R-\sqrt{R^2+1})}{2-P(1+R+\sqrt{R^2+1})}} = 0.388$$

则 $\Delta T_{m,1} = F_T \cdot \Delta T_{m,cc} = 23.71°C$

$Q_1 = K_1 A \Delta T_{m,1} = 11855 W < 22650 W$ ×

换热器 2: 单壳程单管程, $K_2 = 500 W/(m^2 \cdot °C)$

$\Delta T_{m,2} = \Delta T_{m,cc} = 61.11°C$

$Q_2 = K_2 A \Delta T_{m,2} = 24444 W > 22650 W$, 可

[变式 6-5] ……

[变式 6-6]

$T_{h1} = 95°C$, $T_{c1} = 15°C$, $T_{c2} = 55°C$

$Q = m_w c_{p,w} (T_{h1} - T_{h2})$

$= m_{air} c_{p,air} (T_{c2} - T_{c1})$

$= 11166.7 W$

$\Rightarrow T_{h2} = 82.98°C$

$95 \to 82.98$

$55 \leftarrow 15$

(1) $K = 180 W/(m^2 \cdot °C)$

$$\Delta T_{m,1} = \frac{67.98 - 40}{\ln \frac{67.98}{40}} = 52.76°C$$

$Q_1 = K A \Delta T_{m,1} = 18993.6 W > Q$

可

(2) $K = 220 W/(m^2 \cdot °C)$

$$R = \frac{95 - 82.98}{55 - 15} = 0.3005$$

$$P = \frac{55 - 15}{95 - 15} = 0.5$$

$$|F_T| = \left| \frac{\sqrt{R^2+1} \ln \frac{1-PR}{1-P}}{(R-1) \ln \frac{2-P(1+R-\sqrt{R^2+1})}{2-P(1+R+\sqrt{R^2+1})}} \right| = 0.97$$

则 $\Delta T_{m,2} = \Delta T_{m,1} F_T$

$= 51.18°C$

$Q_2 = K A \Delta T_{m,2} = 22519.2 W > Q$

可

[6-7] $m_b = 1.25\ kg/s$, $C_b = 1705\ J/(kg\cdot K)$

$T_{h1} = 350K$, $T_{h2} = 300K$.

$T_{c1} = 290K$, $T_{c2} = 320K$

$a_w = 850\ W/(m^2\cdot K)$, $\alpha_b = 1700\ W/(m^2\cdot K)$

$d_o = 0.025\ m$, $d_i = 0.020m$, $\frac{A_o}{A_i} = \frac{d_o}{d_i} = 1.25$

$$Q = m_b C_b (T_{h1} - T_{h2}) = m_w C_w (T_{c2} - T_{c1}) \Rightarrow \begin{cases} Q = 106563W \\ m_c = 0.850\ kg/s \end{cases}$$

350 → 300

320 ← 290 $\Delta T_m = 18.20K$

$$\frac{1}{K_o} = \frac{1}{a_o} + \frac{1}{a_i}\frac{A_o}{A_i} \Rightarrow K_o = 485.7\ W/(m^2\cdot K)$$

$$A_o = \frac{Q}{K_o \Delta T_m} = \frac{106563}{485.7\times 18.20} = 12.05\ m^2$$

$$L = \frac{A_o}{\pi d_o} = 153.5m$$

[6-8]

[变式6-7]

$$Q = m_o C_{p,o} (T_{h1} - T_{h2}) = m_w C_{p,w} (T_{c2} - T_{c1})$$

$$\Rightarrow \begin{cases} Q = 80000\ W \\ m_w = 0.766\ kg/s \end{cases}$$

~~120°C~~

120°C → 70°C

50°C ← 25°C

$$\Delta T_m = \frac{\cancel{50-25}\ 70-45}{\ln\frac{70}{45}} = \cancel{36.07}\ 56.58\ ^\circ C$$

$$Q = K A_o \Delta T_m$$

$\cancel{\Rightarrow A_o}$

$$\frac{1}{K_o} = \frac{1}{a_o} + \frac{1}{a_i}\frac{A_o}{A_i} \Rightarrow K_o = 371.43\ W/(m^2\cdot{}^\circ C)$$

$$\Rightarrow A_o = \cancel{5.97m}\ 3.803m^2$$

$$L = \frac{A_o}{\pi d_o} = \cancel{59.38m}\ 37.84\ m$$

化工原理 II

第7章 蒸发

「蒸发水量的计算」

由单位时间内随原料进入蒸发器的溶质质量与随完成液离开蒸发器的溶质质量相等得

$$\begin{cases} Fx_0 = Lx_1 \\ F = L + W \end{cases} \Rightarrow \begin{cases} \text{水的蒸发量}: W = F\left(1-\frac{x_0}{x_1}\right) \\ \text{完成液量}: L = F - W = F\frac{x_0}{x_1} \end{cases}$$

「蒸发器热负荷的确定」

设加热蒸气的冷凝液在饱和温度下排出，

则蒸发器的热量衡算为 $DH + Fh_0 = WH' + (F-W)h_1 + Dh_w + Q_L$

$$\Rightarrow D = \frac{Fc_0(t_1-t_0) + Wr' + Q_L}{r}$$

其中 $H - c_wT = r$，$H' - c_wt_1 = r'$

→ 若原料在沸点下进入蒸发器，即 $t_0 = t_1$，且蒸发器的热损失可以忽略，

$$\Rightarrow \frac{D}{W} = \frac{r'}{r}$$

「传热面积的计算」

$$\begin{cases} Q = KA\Delta t_m \\ Q = D(H - h_w) = Dr \end{cases}$$

传热平均温度差 $\Delta t_m = T - t_1$

由温差分配公式 $\Delta t_i \propto \frac{1}{K_i}$ 的前提：

→ 各效传热量近似相等

→ 稀溶液、进料为沸点… 无额外蒸气引出

若各效传热量差异较大，则需迭代计算

「多效蒸发的经济性」

1kg 蒸气可蒸发的水分量：$E = \frac{W}{D}$

「单效蒸发物料衡算」

$$\begin{cases} F = L + V \\ Fx_F = Lx_L \end{cases}$$

$$\Rightarrow V = F\left(1 - \frac{x_F}{x_L}\right)$$

「单效蒸发热量衡算」

加热蒸气放热量 = 原料液升温 + 水分汽化 + 热损失

$$DH_s + Fh_F = Lh_L + VH_V + Q_{loss}$$

若忽略溶液的浓缩热，且溶液的比热容为常数：

$$Dr_s = Fc_{p0}(t_1 - t_0) + Vr' + Q_{loss}$$

「传热方程」

$$Q = KA\Delta t_{eff} = Dr_s$$

「有效传热温差」

$\Delta t_{eff} = T_s - t_1$ = 加热蒸气的饱和温度 − 溶液的沸点

其中 $t_1 = T_v' + \Delta' + \Delta'' + \Delta'''$

沸点↑ 静压力 管路流动阻力

「静压力引起的温度差损失」

$\Delta'' = T_v(p_m) - T_v(p')$

其中 $p_m = p' + \frac{\rho g h}{2}$，$p'$ 为液面处压力

「蒸气经济」 $= \frac{V}{D}$

「多效蒸气温差分配」

各效有效温差 $\Delta t_i \propto \frac{1}{K_i}$

总有效温差 $\sum \Delta t_i = T_s - T_k$

「多效并流蒸发物料衡算」

第 i 效浓度：$x_i = \dfrac{Fx_F}{F - \sum_{j=1}^{i} V_j}$

化原Ⅱ 第7章习题

【7-1】 —— 传热有效温差

$x = 30\%$

蒸气室压力 $p' = 60kPa$ 　　$P_{abs} = P_{gauge} + P_{atm}$

$h = 2m$, $\rho = 1280kg/m^3$

$P_s = 1MPa \Rightarrow P_{s,abs} = 1.0 + 0.1013 = 1.1013MPa$

查饱和水蒸气表得：$T_s = 184.1K$

而蒸发室压力下纯水的饱和温度 $T_v' = 85.9K$

在 $p' = 60kPa$ 时，$\Delta' = 23.1$

$P_m = p' + \frac{\rho g h}{2} = 72.56kPa \Rightarrow T_v(P_m) = 90.9K$

$\Delta'' = T_v(P_m) - T_v' = 5.0K$

则 $t_1 = T_v' + \Delta' + \Delta'' = 114.0K$

$\Delta t_{eff} = T_s - t_1 = 70.1K$

【7-2】 — 完成液浓度计算

已知 $F = 1600kg/h$, $A = 85m^2$, $x_F = 0.10$.

$t_0 = 30K$, $P_{蒸} = 100kPa$, $P_{蒸} = 200kPa$

$\Delta t_{eff} = 12K$, $K = 900W/(m^2 \cdot K)$

查饱和水蒸气表：

$P = 200kPa$ 时，$T_s = 120.2K$, $r_s = 2204.6kJ/kg$

$P = 100kPa$ 时，$T_v' = 99.6K$, $r' = 2259.5kJ/kg$

由有效温差定义 $\Delta t_{eff} = T_s - t_1 = 12K$，则 $t_1 = 108.2K$

则传热量 $Q = KA\Delta t_{eff} = 918kW$

加热蒸汽量 $D = \frac{Q}{r_s} = \frac{918}{2204.6} \times 3600 = 1500kg/h$

而有效热量 $Q_{net} = Q \times 95\% = 872.1kW$

由热量衡算式 $Q_{net} = FC_{p0}(t_1 - t_0) + Vr'$

$\Rightarrow V = 1184.6kg/h$

完成液量 $L = F - V = 415.4kg/h$，而 $Fx_F = Lx_L \Rightarrow x_L = 38.5\%$

【7-3】 —传热面积与加热蒸气经济性

$P_s = 111.3kPa$, $T_s = 111.4K$, $r_s = 2226kJ/kg$

$P_蒸' = 34.3kPa$, $T_v' = 72.0K$, $r' = 2328kJ/kg$

由物料衡算 $Fx_F = (F - V)x_L$

$\Rightarrow V = 3333.3kg/h$, $L = 1666.7kg/h$

查杜林直线，在 34.3kPa 下，$\Delta' = 16.0K$

则 $t_1 = T_v' + \Delta' = 88.0K$, $\Delta t_{eff} = T_s - t_1 = 23.4K$

当 $x = 10\%$ 时，$C_p = 3.95$，当 $x = 30\%$ 时，$C_p = 3.48$，则取 $C_p = 3.8kJ/(kg \cdot K)$

$0.95Dr_s = FC_p(t_1 - t_0) + Vr' \Rightarrow D = 4010kg/h$

$A = \frac{Dr_s}{K\Delta t_{eff}} = 53.1m^2$，蒸气经济 $= \frac{V}{D} = 0.83$

【变式 7-1】

$p' = 50kPa$

$h = 1.5m$, $\rho = 1150kg/m^3$

$P_s = 0.6MPa$, $\Delta' = 18K$

$P_{s,abs} = 0.7013MPa$

$p' = 50kPa$ 时，$T_v' = 81.2K$

$P_m = p' + \frac{\rho g h}{2} = 58.46kPa$

$T_v(P_m) = 84.7K$, $\Delta'' = T_v(P_m) - T_v' = 3.5K$

$t_1 = T_v' + \Delta' + \Delta'' = 102.7K$

则 $\Delta t_{eff} = T_s - t_1 = 62.3K$

【变式 7-2】

当 $p = 80kPa$ 时，$T_v' = 93.2K$; $r' = 2275.3kJ/kg$

当 $p = 150kPa$ 时，$T_s = 111.4K$, $r_s = 2226kJ/kg$

$\Delta t_{eff} = T_s - t_1$

$\Rightarrow t_1 = 101.4K$

$Q = KA\Delta t_{eff} = 660kW$

$Q_{net} = 96\% Q = 633.6kW$

$= FC_p(t_1 - t_0) + Vr'$

$\Rightarrow V = ~~1000.5kg/h~~ \ 878kg/h$

由 $Fx_F = Lx_L = (F - V)x_L$

$\Rightarrow x_L = ~~48.12\%~~ \ 29.8\%$

「变式 7-3」

$P_s = 0.3 + 0.1013 = 0.4013\,MPa$，$T_s = 143.6K$，$r_s = 2134\,kJ/kg$

$P' = 40kPa$，$T_v' = 75.9K$，$r' = 2319\,kJ/kg$

由物料衡算：$Fx_F = Lx_L$

$\Rightarrow L = 1000kg/h$，$V = F - L = 3000kg/h$

由热量衡算：$0.94Dr_s = FC_p(t_1 - t_0) + Vr'$

由传热方程：$Q = KA\Delta t_{eff} = Dr_s$

$$\left\{\begin{array}{l}\text{而 } \Delta t_{eff} = T_s - t_1 = 67.7K \\ t_1 = T_v' + \Delta' + \Delta'' + \Delta''' = T_v' = 75.9K\end{array}\right.$$ ，取 $C_p = 4.0kJ/(kg\cdot K)$

$$13090\overline{)47139}^{\;3}$$
$$39270$$
$$80690$$

$$\Rightarrow \left\{\begin{array}{l}\text{去 } A = 17.5m^2 \\ D = 3595\,kg/h\end{array}\right.$$

蒸发经济 $= \frac{V}{D} = 0.83$

「7-4」蒸发水量与传热系数

$p' = 101.3kPa$，$T_v' = 100K$，$r' = 2258.4\,kJ/kg$

$T_s = 110K$，$P_s = 143.31kPa$，$r_s = 2232.0kJ/kg$

(1) $Fx_F = (F-V)x_L \Rightarrow V = 1500kg/h$，而 $t_1 = T_v' + \Delta' = 100K$，$\Delta t_{eff} = 10K$

由传热方程 $Q = KA\Delta t_{eff} = Dr_s = FC_p(t_1 - t_0) + Vr'$

$\Rightarrow K = 1.88\,kW/(m^2\cdot K)$

(2) K 与 Δt_{eff} 不变，则 $Dr_s = 1310.36\,kW$

$= F'C_p(t_1 - t_0) + (F' - L')r'$

而 $F'x_F = L'x_L'$

$\Rightarrow x_L' = 2.429\%$

「变式 7-4」 查表得

(1) $T_s = 120K$，$P_s = 198.64kPa$，$r_s = 2205.2\,kJ/kg$

$p' = 1atm = 1.013kPa$，$T_v' = 100K$，$r' = 2258.4\,kJ/kg$

$Fx_F = Lx_L \Rightarrow L = F\frac{x_F}{x_L} = 1875$ ~~kJ/kg~~ kg/h，$V = F - L =$ ~~1125 kJ/kg~~ 1125 kg/h

因稀溶液的沸点升高可忽略，则 $t_1 = T_v' + \Delta' = 100K$

由热量衡算：$Q = FC_p(t_1 - t_0) + Vr' =$ ~~33012~~ $3301200\,kJ/h$

$= 917000\,W$

$\Delta t_{eff} = T_s - t_1 = 20K$

则 $Q = KA\Delta t_{eff} \Rightarrow K = 917\,W/(m^2\cdot K)$

(2) Δt_{eff} 不变，则 Q 不变

$\Rightarrow V' = 1012.75$ ~~kJ/kg~~ kg/h

$\Rightarrow L' = 2987.25\,kg/h$

则 $x_L' = \frac{F}{L'}x_F = 6.7\%$

「7-5」各效溶液沸点.　　各效温差与传热系数成反比：$\Delta t_i \propto \frac{1}{K_i}$

$T_s = 121K$

$P' = \cancel{26.1k}$ $26.1kPa$，$T' = 64K$

$\Delta T_{总} = T_s - T' = 57K$

而 $\Delta t_1 : \Delta t_2 : \Delta t_3 = \frac{1}{K_1} : \frac{1}{K_2} : \frac{1}{K_3}$

$\Rightarrow \Delta t_1 = 12.88K$；$\Delta t_2 = 18.37K$；$\Delta t_3 = 25.75K$

$$\begin{cases} t_1 = T_s - \Delta t_1 = 108.12K \\ t_2 = t_1 - \Delta t_2 = 89.75K \\ t_3 = t_2 - \Delta t_3 = 64.0K \end{cases}$$

「变式 7-5」

$T_s = 130K$，$t_3 = 55K$

$\Delta T_{总} = 75K$

$\sum \frac{1}{K_i} = 1.806 \times 10^{-3}$

则 $\Delta t_1 = \Delta T_{总} \times \frac{1}{1.806 \times 10^{-3} \times 2400} = 17.3K$

$\Delta t_2 = 23.1K$

$\Delta t_3 = 34.6K$

则 $\begin{cases} t_1 = T_s - \Delta t_1 = 112.7K \\ t_2 = t_1 - \Delta t_2 = 89.6K \\ t_3 = t_2 - \Delta t_3 = 55K \end{cases}$

「7-6」双效并流蒸发 —传热面积

试差法

$T_s = 120K$，$r_s = 2205kJ/kg$

$T_{v_2} = 53K$，

$V_{总} = F(1 - \frac{x_F}{x_L}) = 1777.8kg/h = V_1 + V_2$

$$\begin{cases} \Delta t_1 = T_s - t_1 \\ \Delta t_2 = t_1 - T_2 \end{cases}$$

设 $t_1 = 91.5K$ $\Rightarrow \Delta t_1 = 28.5K$，$\Delta t_2 = 38.5K$

第二效热量衡算　$V_1 r_{v_1} = V_2 r_{v_2} + (F - V_1) C_p (t_1 - T_2)$

……　精校得 $A = 9.526m^2$

「变式 7-6」

$T_s = 115K$，$r_s = 2217kJ/kg$

$T_2 = 50K$

设 $t_1 = 88K$

试差求平均 $A = 15.6m^2$

「7-7」三效蒸发 —总蒸发量与各效浓度

$F = 2000kg/h$　　$V_{总} = F(1 - \frac{x_F}{x_L}) = 1500kg/h$

$V_2 = 1.15V_1$，$V_3 = 1.3V_1$

~~$Fx_F = V_{总} x_{V_3} \Rightarrow V_3 = 500kg/h$~~

$\Rightarrow$ ~~$V_1 = 384.6kg/h$~~

~~$V_2 = 442.3kg/h$~~

$$\Rightarrow \begin{cases} V_1 = 434.78kg/h \\ V_2 = 500kg/h \\ V_3 = 565.22kg/h \end{cases} \Rightarrow \begin{cases} x_1 = 12.78\% \\ x_2 = 18.78\% \\ x_3 = 40.0\% \end{cases}$$

「7-8」「变式 7-8」上机

「变式 7-7」

$F = 3000kg/h$

$$\begin{cases} V_1 = 0.35V \\ V_2 = 0.35V \\ V_3 = 0.3V \end{cases}$$

$Fx_F = Lx_L \Rightarrow V = F(1 - \frac{x_F}{x_L}) = 2200kg/h$

$$\Rightarrow \begin{cases} V_1 = 770kg/h \\ V_2 = 770kg/h \\ V_3 = 660kg/h \end{cases}$$

$Fx_F = (F - V_1)x_{V_1} \Rightarrow \begin{cases} x_{V_1} = 10.76\% \\ x_{V_2} = 16.44\% \\ x_{V_3} = 30\% \end{cases}$

化工原理II

第8章 质量传递基础

[浓度表示法及换算]

质量浓度：$\rho_A = \frac{m_A}{V}$ (kg/m^3)

摩尔浓度：$C_A = \frac{n_A}{V}$ ($kmol/m^3$)

$C_A = \frac{\rho_A}{M_A}$

摩尔分数：液相 $x_A = \frac{C_A}{C}$，气相 $y_A = \frac{p_A}{P}$

分压与浓度 (ideal gas)：$p_A = C_A RT$

总摩尔浓度(气相)：$C = \frac{P}{RT}$

[分子扩散与费克定律]

费克第一定律

$$J_A = -D_{AB}\frac{dC_A}{dz}$$

D_{AB} → A在B中的扩散系数 m^2/s 为物性

对于一维稳态：$J_A = D_{AB}\frac{C_{A1}-C_{A2}}{\delta}$

总通量 $N_A = J_A + x_A N = -D_{AB}\frac{dC_A}{dz} + x_A(N_A+N_B)$

[等摩尔反向扩散]

$N_A + N_B = 0$，$N_A = \frac{D_{AB}}{\delta}(C_{A1}-C_{A2}) = \frac{D_{AB}}{RT\delta}(p_{A1}-p_{A2})$

[单组分A通过停滞组分B的扩散]

条件：$N_B = 0$

$$N_A = \frac{D_{AB}P}{RT\delta}\ln\frac{P-p_{A2}}{P-p_{A1}} = \frac{D_{AB}P}{RT\delta}\frac{p_{A1}-p_{A2}}{p_{Bm}}$$

where $p_{Bm} = \frac{p_{B2}-p_{B1}}{\ln\frac{p_{B2}}{p_{B1}}}$

[沿程分压]

$$\frac{P-p_A(z)}{P-p_{A1}} = \left(\frac{P-p_{A2}}{P-p_{A1}}\right)^{\frac{z}{\delta}}$$

[对流传质与传质系数]

Sherwood数：$Sh = \frac{kd}{D_{AB}}$，k：传质系数

Schmidt数：$Sc = \frac{\nu}{D_{AB}}$ ⇐ $Re = \frac{du}{\nu}$

(管内湍流) Gilliland-Sherwood关联式：

$$Sh = 0.023Re^{0.83}Sc^{0.44}$$

传质系数：$k = Sh\cdot\frac{D_{AB}}{d}$

$$D \propto \frac{T^{1.5}}{P}$$

[8-1]

$N_A = -N_B$

$N_A = D_{AB}\frac{C_{A1}-C_{A2}}{\delta}$

$\delta = 25mm = 0.025m$

$T = 25°C = 298K$，$P = 1atm = 101.3kPa$

H_2与Cl_2以相同物质的量扩散混合，

一端为纯H_2 ($p_{A1} = P$)，另一端为纯Cl_2 ($p_{A2} = 0$)

$\Delta p_A = \frac{P}{2} = 50.7kPa$

$N_A = \frac{D_{AB}\Delta p_A}{RT\delta} = 4.30\times10^{-5}\ kmol/(m^2\cdot s)$

[变式8-1]

$N_{N_2} = \frac{D_{AB}\Delta p_{N_2}}{RT\delta} = 2.05\times10^{-6}\ kmol/(m^2\cdot s)$

[8-2]

NH_3-N_2扩散体系，$N_B = 0$

$d = 0.0244m$，$\delta = 0.61m$，$P = 101.3kPa$，$T = 298K$

$p_{A1} = 20kPa$，$p_{A2} = 6.67kPa$，$D_{AB} = 2.30\times10^{-5}\ m^2/s$

(1) 惰性组分N_2在两端的分数

$p_{B1} = 81.3kPa$，$p_{B2} = 94.63kPa$

⇒ $p_{Bm} = 87.87kPa$，则 $N_A = \frac{D_{AB}P}{RT\delta}\frac{p_{A1}-p_{A2}}{p_{Bm}} = 2.34\times10^{-7}\ kmol/(m^2\cdot s)$

(2) $\frac{P-p(z)}{P-p_{A1}} = \left(\frac{P-p_{A2}}{P-p_{A1}}\right)^{\frac{z}{\delta}}$ ⇒ $p_A = 13.5kPa$

指数分布

[变式8-2]

$d = 0.02m$，$\delta = 0.50m$

$P = 101.3kPa$，$T = 303K$

$p_{A1} = 15kPa$，$p_{A2} = 5kPa$，$D_{AB} = 1.22\times10^{-5}\ m^2/s$

$p_{B1} = \cancel{288}\ 86.3\ kPa$，$p_{B2} = \cancel{298}\ 96.3\ kPa$

$p_{Bm} = \cancel{\frac{298-288}{\ln\frac{298}{288}} = 292.97kPa}\ \frac{10}{\ln\frac{96.3}{86.3}} = 91.21\ kPa$

$N_{SO_2} = \frac{D_{AB}P}{RT\delta}\frac{p_{A1}-p_{A2}}{p_{Bm}} = 1.076\times10^{-7}\ kmol/(m^2\cdot s)$

~~$= 3.35\times10^{-8}\ kmol/(m^2\cdot s)$~~

$\frac{P-p_A}{P-p_{A1}} = \left(\frac{P-p_{A2}}{P-p_{A1}}\right)^{\frac{z}{\delta}}$ ⇒ $p_A = 10.14kPa$

[8-3] ?

等摩尔反向扩散 $N_A = -N_B$

$p = 101.3\ kPa$, $T = 273K$, $y_{A_1} = 0.20$, $y_{A_2} = 0.02$, $\delta = 3m$

$D = 1.44 \times 10^{-5}\ m^2/s$

总摩尔浓度 $c = \frac{p}{RT} = \frac{101.3 \times \cancel{10^3}}{8.314 \times 273} = 0.04463\ kmol/m^3$

A点处 CO_2 的摩尔浓度 $C_{A1} = cy_{A1} = 0.008926\ kmol/m^3$

(1) CO_2 的扩散通量

$$N_A = \frac{Dc(y_{A_1} - y_{A_2})}{\delta} = \cancel{3.86 \times 10^{-8}\ kmol/(m^2 \cdot s)}\ 3.86 \times 10^{-8}\ kmol/(m^2 \cdot s)$$

则 N_2 扩散的速度: $v = \frac{N_A}{C_{A1}} = \cancel{4.32 \times 10^{-6}\ m/s}\ 4.32 \times 10^{-6}\ m/s$

$\cancel{= 0.0156\ m/h} = 0.0156\ m/h$

(2) $N_A = N_A^{diff} + C_{A_1} v$

$$N_A = \frac{Dc}{\delta} \ln \frac{1 - y_{A_2}}{1 - y_{A_1}}$$

$$= 1.735 \times 10^{-4}\ kmol/(m^2 \cdot h)$$

[变式8-3] $O_2 - N_2$

$T = 298K$,

$y_{AO_2} = 0.25$, $y_{BO_2} = 0.05$

$c = \frac{p}{RT} = 0.04089\ kmol/m^3$

$N_A = \frac{Dc}{\delta} \ln \frac{1 - y_{A_2}}{1 - y_{A_1}} = 9.13 \times 10^{-8}\ kmol/(m^2 \cdot s)$

$N_A^{diff} = \frac{D_{AB}\, c\,(y_{AO_2} - y_{BO_2})}{\delta} = 8.38 \times 10^{-8}\ kmol/(m^2 \cdot s)$, $v = \frac{N_A^{diff}}{p y_{AO_2}} = 0.0295\ m/h$

[8-4] ...

[8-5]

练习八

1. 已得 $C_{Rc}=\frac{m_c P_{pc}}{m_h C_{ph}}$ ，$NTU=\frac{KA}{m_c C_{pc}} \rightarrow +\infty$

逆流：$\varepsilon=\frac{t_2-t_1}{T_1-t_1}=\frac{1-\exp((1-C_{Rc})NTU)}{C_{Rc}-\exp((1-C_{Rc})NTU)}$ 　　并流：$\varepsilon=\frac{t_2-t_1}{T_1-t_1}=\frac{1-\exp(-(1+C_{Rc})NTU)}{1+C_{Rc}}$

(1) $C_{Rc}=2$ ，则 $\varepsilon=\frac{t_2-50}{150}=\frac{1}{3} \Rightarrow t_2=100^\circ C$

(2) $C_{Rc}=2$，则 $\varepsilon=\frac{t_2-50}{150}=\frac{1}{2} \Rightarrow t_2=125^\circ C$

(3) $C_{Rc}=0.5$，则 $\varepsilon=\frac{t_2-50}{150}=1 \Rightarrow t_2=200^\circ C$

2. $N=269$，$L=3m$，$d=0.025\,m$

$\dot m_a=8000\,kg/h$，~~若为理想，$\Delta t_m=\frac{10-xx}{\ln\frac{x}{x}}$~~

默认为逆流，则 $Q=\dot m_a C_p(t_2-t_1)=KA\Delta t_m \approx \alpha A\Delta t_m \Rightarrow \Delta t_m=73.05$

$=\dot m_g \cdot C_p(T_1-T_2)$

$120^\circ C \rightarrow T_2$

$110^\circ C \leftarrow 10^\circ C$ 　　$\Rightarrow \Delta t_m=\frac{T_2-20}{\ln\frac{T_2-10}{10}} \Rightarrow T_2=$

$$\begin{cases} Nu=\frac{\alpha d}{\lambda}=0.023Re^{0.8}Pr^{0.4} \Rightarrow \alpha=183.29\ W/(m^2\cdot K) \\ Re=\frac{du\rho}{\mu}=\frac{d\rho}{\mu}\times\frac{\dot m_a}{\rho\times\frac{\pi}{4}d^2N}=75355>10^4 \\ Pr=\frac{c_p\mu}{\lambda}=0.707>0.7 \end{cases}$$

，而 $A=NL\times\pi d=63.38m^2$

3.

$130^\circ C \rightarrow 70^\circ C$

$90^\circ C \leftarrow 30^\circ C$

(1) $Q=\dot m_h C_{ph}(T_{h_1}-T_{h_2})=KA\Delta t_m$

$=\dot m_c C_{p,c}(T_{c_2}-T_{c_1})$ ，$C_{p,c}=4.187kJ/(kg\cdot K)$

$\frac{1}{K}=\frac{1}{\alpha_h}+\frac{1}{\alpha_c} \Rightarrow K=40W/(m^2\cdot K)$

$\Rightarrow \dot m_c=1809.17\,kg/h$

此时 $\dot m_c'=3618.34\,kg/h$

$\Rightarrow T_{c_2}'=$

2. $N=269$, $L=3m$, $d=0.025m$

$A_{交} = N \times \pi d L = 63.38 m^2$

$\therefore Q = \dot{m}_a C_{p.a} (T_{h1} - T_{h2}) = KA\Delta t_m$

$= 224.44 kW$

~~$C_{Rc} =$~~ $NTU_c = \frac{KA}{\dot{m}_a C_{p.a}} = 2.93 \times 10^{-2} A$

~~C_{Rc}~~ 而 $C_{Rc} = \frac{\dot{m}_a C_{p.a}}{m} = 0$

则 $\varepsilon = \frac{t_2 - t_1}{T_{h1} - t_1} = \frac{1-\exp((1-C_{Rc})NTU)}{C_{Rc} - \exp((1-C_{Rc})NTU)}$

$= \frac{1-e^{NTU}}{-e^{NTU}}$

将 $A = A_{交} = 63.38 m^2$ 代入解得

$t_2 = 102.8°C$

$\bar{t}_a = \frac{t_2+t_1}{2} = 56.4°C$

则 $\bar{t}_w - \bar{t}_a = \frac{Q}{A\alpha}$

$\Rightarrow \bar{t}_w = 106.4°C$

则更接近蒸汽侧

$Nu = \frac{\alpha d}{\lambda} = 0.023 Re^{0.8} Pr^{0.4}$

而 $Re = \frac{du\rho}{\mu} = \frac{d\rho}{\mu} \times \frac{\dot{m}_a}{\rho \times \frac{\pi}{4} d^2 \times N} = 2.09 \times 10^4 > 10^4$

$Pr = \frac{C_p \mu}{\lambda} = 0.707 > 0.7$

$\Rightarrow \alpha = 65.70 W/(m^2 \cdot K) \approx K$

则有 $\frac{110-10}{120-10} = 1 - e^{-NTU} \Rightarrow A_{需} = 81.84m^2 > A_{交}$, 则不合算

3.

化工原理 II　　吸收应在低温、高压下进行
第9章　气体吸收

「气液相平衡与亨利定律」

亨利定律 $\begin{cases} P_A^* = E x_A = \frac{C_A}{H} \\ y_A^* = m x_A \end{cases}$

P_A^*：与液相组成 x_A 平衡的气相分压
E：亨利系数
$m = \frac{E}{P}$，为相平衡常数
H：溶解度系数

「吸收与解吸」

吸收，$y > y^*$，
解吸，$y < y^*$

气相总推动力 $\Delta y = |y - y^*|$
液相总推动力 $\Delta x = |x - x^*|$

「双膜理论」

Whitman 双膜模型：气相主体 $\xrightarrow{气膜}$ 相界面（$y_i = m x_i$）$\xrightarrow{液膜}$ 液相主体

传质速率 $\begin{cases} N_A = k_y (y - y_i) \\ N_A = k_x (x_i - x) \end{cases}$

其中 k 为传质分系数

「总传质系数与阻力叠加」

对流传质方程：

由 $\begin{cases} N_A = k_y (y - y_i) = k_x (x_i - x) \\ y_i = m x_i \end{cases}$

$\Rightarrow \begin{cases} \frac{1}{K_y} = \frac{1}{k_y} + \frac{m}{k_x} \\ \frac{1}{K_x} = \frac{1}{m k_y} + \frac{1}{k_x} \end{cases}$

体积传质系数 $K_y a$
a：气液接触面积（m^2/m^3）

「物料衡算与操作线」

$G(y_b - y_a) = L(x_b - x_a)$
G：惰性气体摩尔流率
L：纯溶剂摩尔流率
a 为塔顶，b 为塔底

操作线方程：$y = \frac{L}{G} x + (y_b - \frac{L}{G} x_b)$

☆吸收因数法 $N_{OG} = \frac{1}{1-S} \ln\left[(1-S)\frac{y_b - y_a^*}{y_a - y_a^*} + S\right]$

$N_{OL} = \frac{1}{1-A} \ln\left[(1-A)\frac{x_b^* - x_a}{x_b^* - x_b} + A\right]$

「填料塔高度计算」

$Z = H_{OG} \cdot N_{OG}$

对数平均推动力法
$N_{OG} = \frac{y_b - y_a}{\Delta y_m}$ 气相总传质单元数

$\Delta y_m = \frac{(y_b - y_b^*) - (y_a - y_a^*)}{\ln \frac{y_b - y_b^*}{y_a - y_a^*}}$

$H_{OG} = \frac{G}{K_y a}$ 气相总传质单元高度

$\eta = \frac{y_b - y_a}{y_b}$

「最小液气比」

当操作线下端与平衡线相交
即 $x_b = x_b^* = \frac{y_b}{m}$ 时，所需液气比最小

$\left(\frac{L}{G}\right)_{min} = \frac{y_b - y_a}{x_b^* - x_a} = \frac{y_b - y_a}{\frac{y_b}{m} - x_a}$

[9-1] T↑ Then H_{OG}, y_a, x_b How?

亨利系数随温度升高而增大，$m = \frac{E}{P}$ ↑
则平衡线变陡，传质推动力↓ ⇒ N_{OG}↑

$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m}{k_x} \Rightarrow K_y \downarrow$，$H_{OG} = \frac{G}{K_y a}$ ↑

则 y_a↑，x_b↓

[9-2]

$P = 101.3\,kPa$，$T = 293\,K$
$100g\ H_2O + 1g\ NH_3$
$P_A^* = 800\,Pa = 0.8\,kPa$

$n_{H_2O} = \frac{100}{18} = 5.56\,mol$

$n_{NH_3} = \frac{1}{17} = 0.0588\,mol$

$x_A = \frac{0.0588}{0.0588 + 5.56} = 0.01046$

(1) $E = \frac{P_A^*}{x_A} = 76.49\,kPa$

(2) $m = \frac{E}{P} = 0.755$

(3) $y_A^* = m x_A = 7.90 \times 10^{-3}$

[变式 9-2]

$P_A^* = 1.2\,kPa$

$x_A = \frac{\frac{3}{64.06}}{\frac{200}{18} + \frac{3}{64.06}} = 4.20 \times 10^{-3}$

$E = \frac{P_A^*}{x_A} = 285.71\,kPa$，$m = \frac{E}{P} = 2.82$

[9-3] $P = 4.0\,MPa$，$T = 32.8°C$

$y_A = 0.05$，$C_A = 10.0\,g/L$
$m = 1.304$，$\rho_L = 950\,kg/m^3 = 0.95\,kg/L$
$M_{MeOH} = 32\,g/mol$，$M_{CO_2} = \cancel{44} 44\,g/mol$

(1) 以 1L 溶液为基准

$n_{CO_2} = \frac{10}{44} = 0.227\,mol$

液相溶液密度 $950 - 10 = 940\,kg/L$

$n_{MeOH} = \frac{940}{32} = 29.375\,mol$

则 $x_A = \frac{n_{CO_2}}{n_{CO_2} + n_{MeOH}} = 7.69 \times 10^{-3}$

液相平衡气相摩尔分数

$y_A^* = m x_A = \cancel{0.001}\ 0.01$

$y_A > y_A^*$，则为吸收

(2) $\Delta y_A = y_A - y_A^* = 0.04$

$\Delta P_A = P \Delta y_A = 0.160\,MPa$

(3) [变式 9-3] P ~~= 2.0MPa, T=25°C, yA=0.08~~ 吸收

~~以 1L 溶液为基准~~

~~$n_{NH_3} = \frac{5-1}{17} = 0.235\,mol$~~

~~$n_{H_2O} = \frac{1}{18} = 0.056\,mol$~~

~~$x_A = 0.808$　$y_A^* = m x_A = 0.606$~~

~~解吸~~

[9-4] $P = 101.3\,kPa,\ T = 293K$

$\begin{cases} K_G = 5\times10^{-6}\ kmol/(m^2\cdot s\cdot kPa) \\ k_L = 1.3\times10^{-4}\ \cancel{kmol/(m^2\cdot s\cdot kPa)}\ m/s \end{cases}$

$y_A = 0.035,\ x_A = 0.009,\ m = 0.8$

(1) $\cancel{k_y = k_x \cdot p}$ $\frac{1}{K_y} = \frac{1}{k_y} + \frac{m}{k_x}$

$k_y = k_G P = 5.065\times10^{-4}\ kmol/(m^2\cdot s)$

$c = \frac{1000}{18} = 55.6\ kmol/m^3$

$k_x = k_L c = 7.228\times10^{-3}\ kmol/(m^2\cdot s)$

气膜阻力 $\frac{1}{k_y} = 1974\ (m^2\cdot s)/kmol$

液膜阻力 $\frac{m}{k_x} = 110.9\ (m^2\cdot s)/kmol$

气膜为主导

总阻力（气相为基准）$\frac{1}{K_y} = 2085\ (m^2\cdot s)/kmol$

再解 $\frac{1}{K_x} = \frac{1}{mk_y} + \frac{1}{k_x} \Rightarrow K_x = 3.84\times10^{-4}\ kmol/(m^2\cdot s\cdot x)$

[变式 9-4] $m = 50$

$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m}{k_x} \Rightarrow K_y = 1.2\times10^{-4}\ kmol/(m^2\cdot s\cdot y)$

$\frac{1}{K_x} = \frac{1}{mk_y} + \frac{1}{k_x} \Rightarrow K_x = 6.1\times10^{-3}\ kmol/(m^2\cdot s\cdot x)$

[9-8] $\eta = 99.5\%,\ y_b = 0.05,\ y = 0.755x$

$G_B = 100\ kmol/h$

$y_a = y_b\times(1-\eta) = 0.00025$

$\Rightarrow m = 0.755,\ x_a = 0$

最小液气比 $x_b^* = \frac{y_b}{m} = 0.06623$

$\cancel{故}\ (\frac{L}{G})_{min} = \frac{y_b - y_a}{x_b^* - x_a} = 0.7512$

(1) 则 $\frac{L}{G} = 1.5\times(\frac{L}{G})_{min} = 1.127$

$L = \frac{L}{G}\times G = 112.7\ kmol/h$

$x_b = \frac{G}{L}(y_b - y_a) = 0.0441$

~~(2) L = 1000~~

吸收量 $G(y_b - y_a)$

$= 4.975\ kmol/h$

(1) 逆流：

$x_b^* = \frac{y_b}{m} = 0.06623$

$L_{min} = \frac{G(y_b - y_a)}{x_b^*}$

$= 75.1\ kmol/h$

$L = L_{min} = 112.7\ kmol/h$

$x_b = \frac{G(y_b - y_a)}{L}$

$= 0.0441$

(2) 并流：$x_b^* = \frac{y_a}{m} = 3.31\times10^{-4}$

$L_{min} = 1.503\times10^{4}\ kmol/h$

$L = L_{min}\times1.5 = 2.254\times10^{4}\ kmol/h$

$x_b = \frac{G(y_b - y_a)}{L} = 2.21\times10^{-4}$

[9-9]

$D = 1.0\,m$

$K_y a = 314\ kmol/(m^3\cdot h\cdot \Delta y)$

$G_s = 94\ kmol/h$

$G' = 119.7\ kmol/(m^2\cdot h)$

$H_{OG} = \frac{G'}{K_y a} = 0.381\,m$

$\Delta Y_m = 0.00333$

$N_{OG} = \frac{Y_b - Y_a}{\Delta Y_m} = 18.9$

$Z = H_{OG}\cdot N_{OG} = 7.2\,m$

[9-11] ……

[9-12] $y = \cancel{0}1.0x,\ y_b = 0.1,\ x_a = 0.01,\ \eta = 80\%$

(1) $y_a = 0.1\times(1-0.80) = 0.02$

$x_b^* = \frac{y_b}{m} = 0.1$

则 $(\frac{L}{G})_{min} = \frac{y_b - y_a}{x_b^* - x_a} = 0.889$, $y_{a,min} = mx_a = 0.01$

$\eta_{max} = \frac{y_b - y_{a,min}}{y_b} = 90\%$

[9-10]

$y_b = 0.02,\ y_a = 0.001$

$P = 101.3\,kPa,\ T = 293K$

$y^* = 1.2x,\ K_y a = 0.0522\ kmol/m^3\cdot s$

$D = 1m$

$G = 0.025\times(1 - y_b) = 0.0245\ kmol/s$

$A_T = \frac{\pi}{4}D^2 = 0.7854$

$G' = G/A_T = \cancel{0.3}\ 0.0312\ kmol/(m^2\cdot s)$，则 $H_{OG} = \frac{G'}{K_y a} = 0.5977\,m$

$\cancel{\Delta Y_b = Y_b - mX_b}$

$\cancel{N_{OG}}$ $N_{OG} = \frac{\Delta Y_b - \Delta Y_a}{\Delta Y_m} = 9.96$

$\begin{cases} Y_b = \frac{0.02}{0.98} = 0.02041 \\ Y_a = \frac{0.001}{0.999} = 0.00100 \end{cases}$

$\Rightarrow Z = 0.598\times9.96$

$= 6.0m$

$X_b^* = \frac{Y_b}{1.2} = 0.01701$

$(\frac{L}{G})_{min} = \frac{Y_b - Y_a}{X_b^* - X_a} = \cancel{0}1.141,\ \frac{L}{G} = 1.2\times1.141 = 1.369$

$X_b = \frac{\cancel{0.0194}\ G(Y_b - Y_a)}{1.369} = 0.01418$